

PMx Weekly Digest: September 05, 2026

Content Source: Full Text

Paper 1: Using Virtual Patients to Evaluate Dosing Strategies for Bispecifics With a Bell

Bibliographic Information

- **Title:** Using Virtual Patients to Evaluate Dosing Strategies for Bispecifics With a Bell-Shaped Efficacy Curve
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- **Journal:** CPT: Pharmacometrics & Systems Pharmacology, 2025
- **DOI:** 10.1002/psp4.70105

Paper Type: POPK_PKPD

Quick Take

This paper presents a semi-mechanistic PK/PD model of teclistamab, a bispecific T-cell engager with a bell-shaped efficacy curve, and uses virtual patient simulations to identify optimal dosing strategies. The model recapitulates the recommended phase II dose and demonstrates that biomarker-based stratification (soluble and membrane-bound BCMA) can refine dose selection, offering a middle ground between population-level and personalized dosing. Pharmacometricians and clinicians involved in bispecific antibody development will find the virtual patient approach and stratification insights valuable.

Executive Summary

The authors develop a semi-mechanistic PK/PD model for teclistamab, a bispecific antibody targeting CD3 and BCMA, which forms a pharmacologically active trimer (PAT) only when bound to both targets. The model incorporates plasma and bone marrow compartments, drug-target interactions, and target turnover. Using sensitivity analysis, they identify key parameters affecting PAT AUC and generate a virtual population of 1200 patients by sampling these parameters from lognormal distributions. Dose-response analysis reveals a bell-shaped efficacy curve, with an optimal weekly dose of 1.8 mg/kg (near-optimal range 1.3–2.5 mg/kg) and biweekly dose of 3 mg/kg (range 2.5–4 mg/kg), consistent with the clinically recommended 1.5 mg/kg Q1W. Stratification by baseline soluble BCMA and membrane-bound BCMA shows that optimal doses vary across subgroups, with high BCMA patients benefiting from higher doses. The study demonstrates the utility of virtual patient modeling and biomarker-based stratification for dose optimization of trimer-based bispecifics, providing a framework for semi-personalized therapy.

Scientific Context & Motivation

Bispecific antibodies that require simultaneous binding to two targets (trimer-based) exhibit a bell-shaped dose-efficacy relationship, where both under- and over-dosing reduce efficacy. This contrasts with Emax models where efficacy plateaus. For teclistamab, a T-cell engager targeting

CD3 and BCMA, the pharmacologically active trimer (PAT) is formed only when the drug binds both targets. Overdosing shifts stoichiometry toward non-productive dimers, leading to loss of efficacy and increased toxicity. Patient heterogeneity in target expression, distribution, and turnover further complicates dose selection. This paper addresses the gap in quantitative frameworks for optimizing dosing of such bispecifics, using a semi-mechanistic model and virtual patient simulations to identify optimal doses and explore biomarker-based stratification.

✂ Methodological Snapshot

A semi-mechanistic PK/PD model of teclistamab with compartments for plasma and bone marrow, tracking free drug, CD3, soluble BCMA, membrane-bound BCMA, and drug-target complexes. The model is parameterized from literature and validated against clinical PK data. Sensitivity analysis identifies key parameters affecting the AUC of the pharmacologically active trimer (PAT). A virtual population of 1200 patients is generated by sampling these parameters from lognormal distributions. Dose-response analysis is performed for weekly and biweekly dosing, and patients are stratified by baseline soluble and membrane-bound BCMA levels.

☒ Structural Model Breakdown

The model includes two compartments: plasma and bone marrow. In each compartment, species include free drug (D), CD3 (T), soluble BCMA (sB), membrane-bound BCMA (mB), and complexes: drug-CD3 (D-T), drug-sBCMA (D-sB), drug-mBCMA (D-mB), and trimer (D-T-mB). The model accounts for drug distribution between plasma and bone marrow, target synthesis and degradation, shedding of mBCMA to sBCMA, and internalization of complexes. The pharmacologically active trimer is formed in the bone marrow when drug binds both CD3 and mBCMA. The model is described in detail in the Supporting Information.

Detailed Methodological Analysis

Modeling Approach

Semi-mechanistic PK/PD model with ordinary differential equations (ODEs) describing drug, CD3, sBCMA, mBCMA dynamics in plasma and bone marrow. The model includes drug distribution between plasma and bone marrow, target binding, internalization, shedding, and synthesis. The pharmacologically active trimer (PAT) is defined as the complex of drug, CD3, and mBCMA. The model is implemented in MATLAB using ode23s solver.

Data Sources

Parameter values from literature: teclistamab PK from MajesTEC-1 trial (Girgis et al.), CD3 and BCMA turnover from published studies, BCMA levels from healthy and MM patients. Model validation against reported PK profiles for 0.72 and 1.5 mg/kg SC doses. Virtual patient parameters sampled from lognormal distributions with means from nominal values and standard deviations based on literature or phenomenological choices.

Estimation Methods

No formal parameter estimation; parameters fixed from literature. Sensitivity analysis performed by perturbing parameters and assessing impact on PAT AUC. Virtual population generated by Monte Carlo sampling.

Model Evaluation

Model validation by comparing simulated PK profiles to reported data. Sensitivity analysis to identify influential parameters. Virtual population analysis to assess dose-response and stratification. No formal goodness-of-fit or bootstrap.

Covariate Analysis

Covariate analysis not performed in traditional sense; instead, patient stratification based on baseline biomarkers (soluble BCMA in plasma, membrane-bound BCMA in bone marrow) to identify subgroups with different optimal doses.

Statistical Rigor Assessment

The virtual population of 1200 patients provides adequate power for median estimates. Sensitivity analysis uses a threshold of 20% change in PAT AUC to classify parameters as sensitive. The dose-response analysis uses median normalized AUC across the population. Stratification uses median cutoffs for biomarkers. No formal statistical testing is performed; the analysis is descriptive. The model does not account for parameter correlations, which may affect variability estimates.

Key Findings

The model reproduces teclistamab PK profiles and predicts a bell-shaped PAT AUC curve. Sensitivity analysis identifies internalization rates of BCMA and CD3, drug distribution into bone marrow, and BCMA shedding as key parameters. Virtual population analysis (n=1200) shows that the median normalized PAT AUC is maximized at 1.8 mg/kg Q1W (near-optimal 1.3–2.5 mg/kg) and 3 mg/kg Q2W (2.5–4 mg/kg), consistent with the RP2D of 1.5 mg/kg Q1W. Stratification by baseline sBCMA and mBCMA reveals that high mBCMA patients require higher doses (1.5–3 mg/kg Q1W) and achieve higher PAT AUC, while low mBCMA patients benefit from lower doses (1–2 mg/kg). Q2W dosing shows greater inter-subpopulation variability than Q1W. The study suggests that biomarker-based stratification could guide dose adjustments, offering a semi-personalized approach.

Clinical & Regulatory Implications

The model predicts that the recommended phase II dose of 1.5 mg/kg Q1W is within the near-optimal range for the average patient. For patients with high baseline mBCMA, higher doses (up to 3 mg/kg Q1W) may be more effective, while low mBCMA patients may benefit from lower doses. Q2W dosing shows greater variability, suggesting that Q1W may be more consistent. Biomarker-based stratification could guide dose adjustments, potentially improving efficacy and reducing toxicity. The approach supports semi-personalized dosing strategies.

Strengths & Limitations

Strengths

- Semi-mechanistic model integrating drug-target interactions in plasma and bone marrow.
- Use of sensitivity analysis to identify key parameters for virtual population generation.
- Virtual population of 1200 patients provides robust statistical power.
- Model recapitulates clinical RP2D, validating the approach.
- Exploration of biomarker-based stratification provides actionable insights.
- Code and data availability on GitHub.

Limitations (Acknowledged by Authors)

- Model is fit-for-purpose and may not capture all biological complexities.
- Assumes lognormal distributions for parameters without empirical justification for all.
- Sensitivity analysis based on AUC of PAT, which may not directly correlate with efficacy.
- Stratification based on median cutoffs may not reflect optimal thresholds.
- Toxicity not explicitly modeled; only efficacy considered.

Limitations (Expert Review)

- The model does not account for target-mediated drug disposition (TMDD) explicitly, though it includes target binding.
- The virtual population sampling assumes independence of parameters, which may not hold biologically.
- The pre-equilibration step may mask initial condition sensitivity.
- The optimal dose range is based on median AUC, ignoring interpatient variability in optimal dose.
- The model does not incorporate immune cell dynamics or tumor burden, which may affect efficacy.
- The stratification analysis uses only two biomarkers; other factors may be relevant.

Generalizability

The framework is generalizable to other trimer-based bispecifics, but requires parameterization for each drug. The approach of using virtual patients and biomarker stratification could be applied to other T-cell engagers, though the specific biomarkers and model structure would need adaptation.

Key Equations

Internalization rate of membrane-bound BCMA

$$k_{\text{int,mB}}$$

Rate at which membrane-bound BCMA is internalized after binding to drug; a key sensitive parameter affecting PAT formation.

Internalization rate of CD3

$$k_{\text{int3}}$$

Rate at which CD3 is internalized after binding to drug; a key sensitive parameter affecting PAT formation.

Drug distribution rate from plasma to bone marrow

$$k_{1bm}$$

Rate constant for drug transfer from plasma to bone marrow compartment; influences drug exposure in the TME.

BCMA shedding rate

$$k_{\text{shed}}$$

Rate at which membrane-bound BCMA is shed to form soluble BCMA; affects soluble BCMA levels and drug sink.

Baseline membrane-bound BCMA in bone marrow

$$mB_{\text{bm0}}$$

Initial concentration of membrane-bound BCMA in the bone marrow; a patient-specific variable used for stratification.

Baseline CD3 in bone marrow

$$CD3_{\text{bm0}}$$

Initial concentration of free CD3 in the bone marrow; a patient-specific variable used for stratification.

Figures & Tables

- **Figure 1:** Schematic representation of the model structure, showing compartments (plasma, bone marrow) and key species (drug, CD3, sBCMA, mBCMA, complexes, trimer).
 - *Significance:* Provides an overview of the model components and interactions, essential for understanding the simulation framework.
- **Figure 2:** Model validation against reported PK profiles. (A) Simulated PK curves for 0.72 and 1.5 mg/kg SC doses; (B) Predicted pharmacologically active trimer (PAT) dynamics for the same doses.
 - *Significance:* Demonstrates that the model reproduces clinical PK data and provides PAT predictions, supporting its use for dose optimization.
- **Figure 3:** Sensitivity analysis results showing the minimal fractional change in each parameter or initial condition required to alter the normalized AUC of the PAT by 20% across doses and schedules.
 - *Significance:* Identifies the most influential parameters (e.g., internalization rates, drug distribution) that drive PAT formation, guiding virtual patient design.

- **Figure 4:** Individual trajectories across the virtual population (1200 patients) at 2 mg/kg. (A) Plasma drug concentration; (B) Pharmacologically active trimer concentration.
 - *Significance:* Illustrates interpatient variability in drug exposure and PAT levels, highlighting the need for personalized dosing considerations.
 - **Figure 5:** Population-level dose-response analysis. (A) Median normalized PAT AUC vs dose for Q1W and Q2W; (B) and (C) Comparison of PAT AUC (blue) and total dimer+trimer AUC (red) for Q1W and Q2W.
 - *Significance:* Reveals the bell-shaped efficacy curve for PAT and the Emax curve for total complexes, demonstrating the need for alternative dose selection strategies for trimer-based bispecifics.
 - **Figure 6:** Four-group stratification based on baseline soluble BCMA (sBp) and membrane-bound BCMA (mBbm) levels. Shows median normalized PAT AUC vs dose for each subgroup for Q1W and Q2W.
 - *Significance:* Demonstrates that optimal doses vary across biomarker-defined subgroups, supporting the concept of biomarker-guided dose selection.
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Code & Reproducibility Assessment

All code is available on GitHub at <https://github.com/jgevertz/teclistamab>. The repository includes MATLAB scripts for model simulation and virtual population analysis.

Supplementary Materials

Supplementary materials include a detailed model description (Supporting Information), Table S1 with nominal parameter values, Table S2 with dose predictions for subpopulations, and additional figures (S1-S8) showing PK profiles, parameter distributions, and stratification results.

Future Directions

Future work could incorporate toxicity models to balance efficacy and safety. The model could be extended to include tumor dynamics and immune cell activation. Validation of stratification predictions with real-world data is needed. Exploration of other biomarkers and optimal stratification thresholds. Application to other trimer-based bispecifics.

Expert Commentary

This paper addresses a critical challenge in dose selection for bispecific antibodies with bell-shaped efficacy curves. The use of virtual patients and sensitivity analysis is a pragmatic approach to account for patient heterogeneity. The finding that biomarker-based stratification can refine dosing is promising, but the model's simplicity and lack of formal parameter estimation limit its predictive power. The authors appropriately acknowledge these limitations. The framework is a valuable addition to the pharmacometrics toolbox, and the open-source code facilitates further exploration. Future studies should incorporate more mechanistic detail and validate predictions clinically.

Bottom Line

This study provides a valuable computational framework for dose optimization of bispecific antibodies with bell-shaped efficacy curves. By combining a semi-mechanistic PK/PD model with virtual patient simulations, it identifies optimal doses for teclistamab and demonstrates that biomarker-based stratification can refine dosing strategies. The approach offers a practical middle ground between population-level and fully personalized dosing, with potential to improve therapeutic outcomes and reduce toxicity. Pharmacometricians should consider integrating such virtual patient analyses into dose selection for trimer-based bispecifics.

☒ Figures

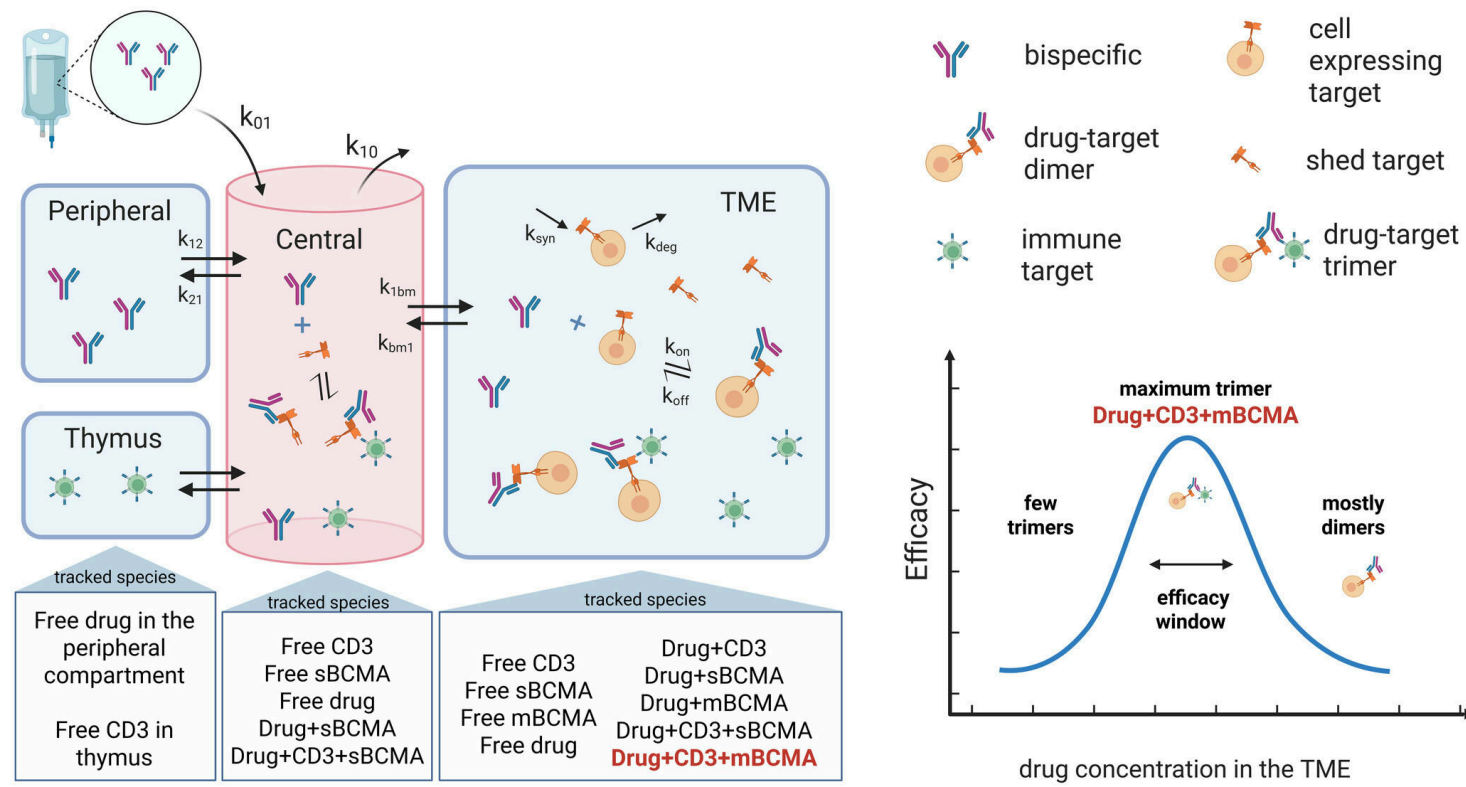


Figure 1: A schematic representation of the key species tracked in the model described in Supporting Information, and the overall objective of maximizing the pharmacological

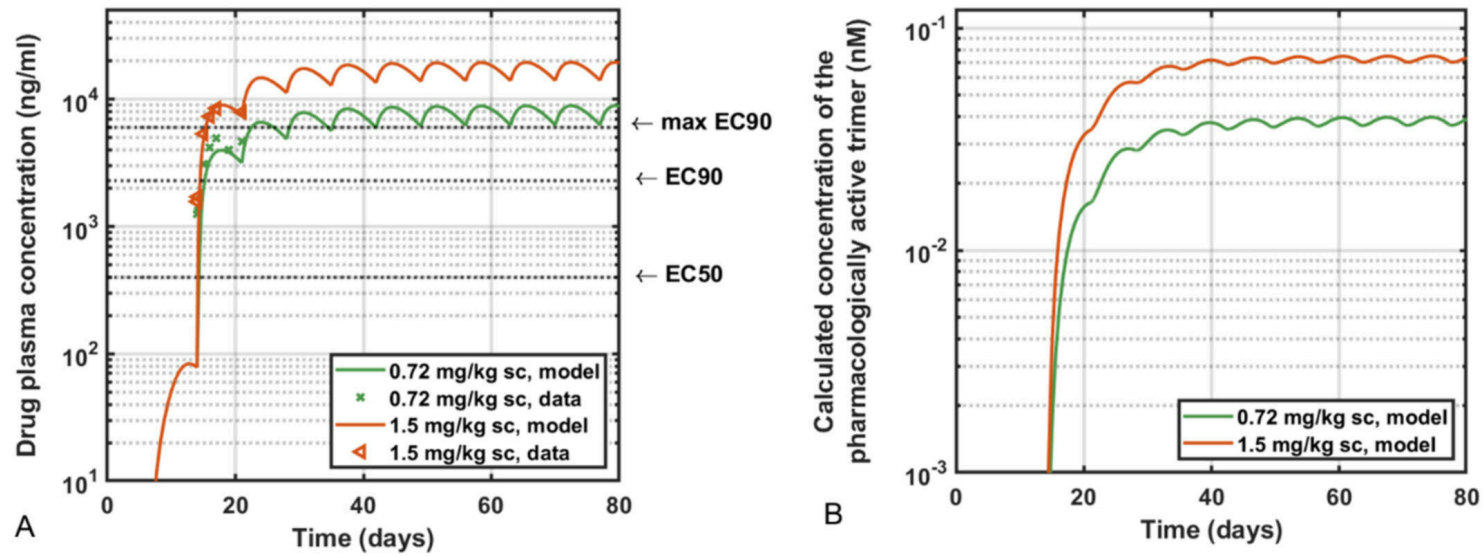


Figure 2: Model validation against reported PK profiles of teclistamab. (A) Simulated PK curves for administration of 0.06 and 0.3 mg/kg loading doses, followed by full do

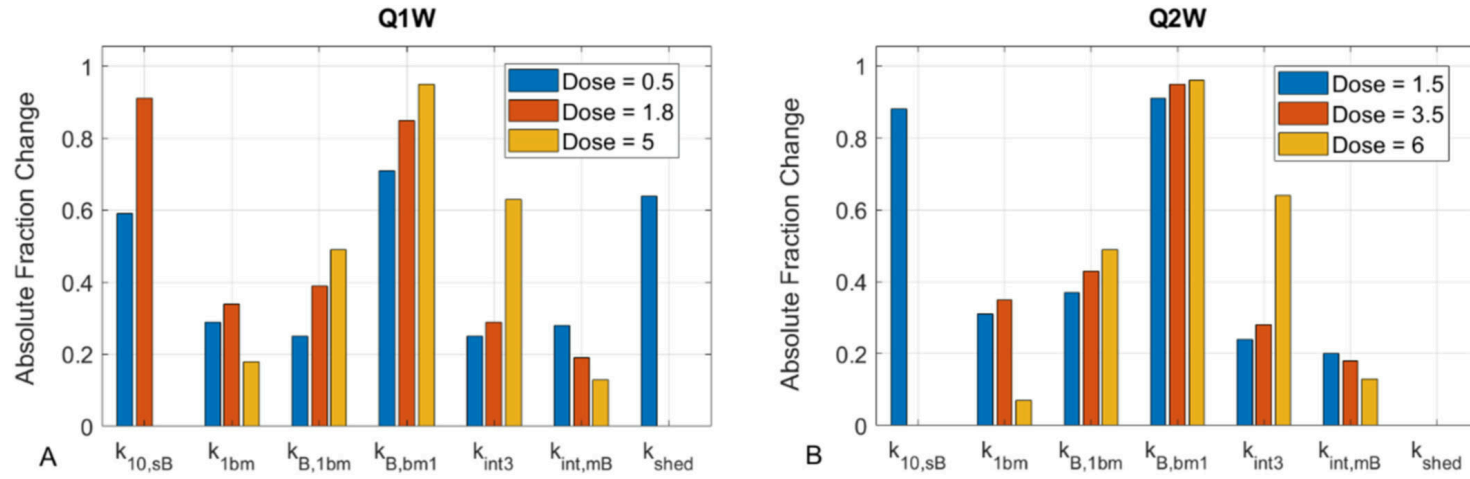


Figure 3: Results of the sensitivity analysis, showing by how much a parameter or a non-zero initial condition needs to be perturbed to change the AUC of the pharmacologic

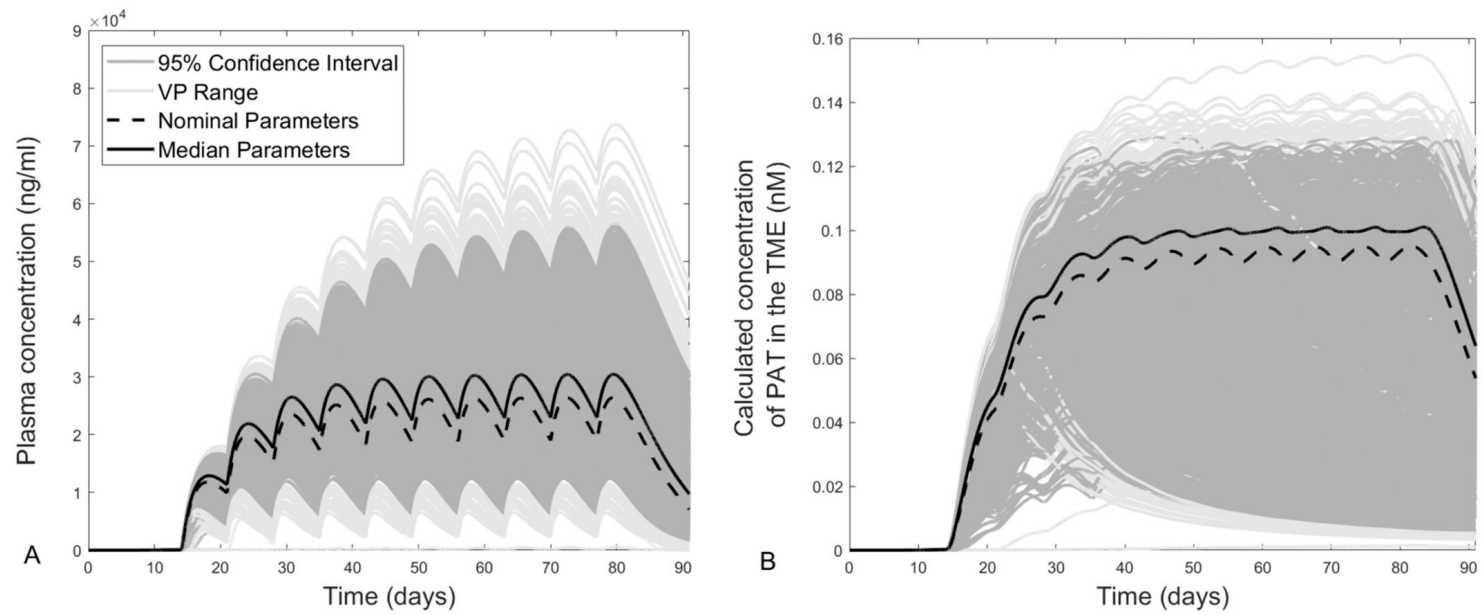


Figure 4: Individual trajectories across the entire virtual population (1200 virtual patients) at a dose of 2 mg/kg. (A) Plasma drug concentration. (B) Pharmacologically a

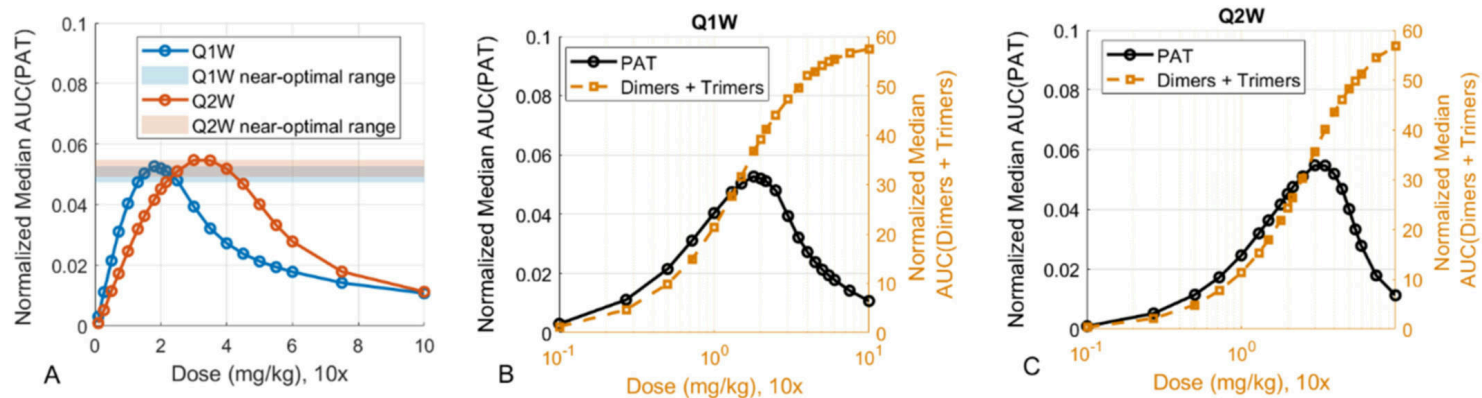


Figure 5: Population-level analysis of doses expected to maximize the normalized AUC of the PAT. (A) Median normalized AUC (across 1200 virtual patients) of the calculated

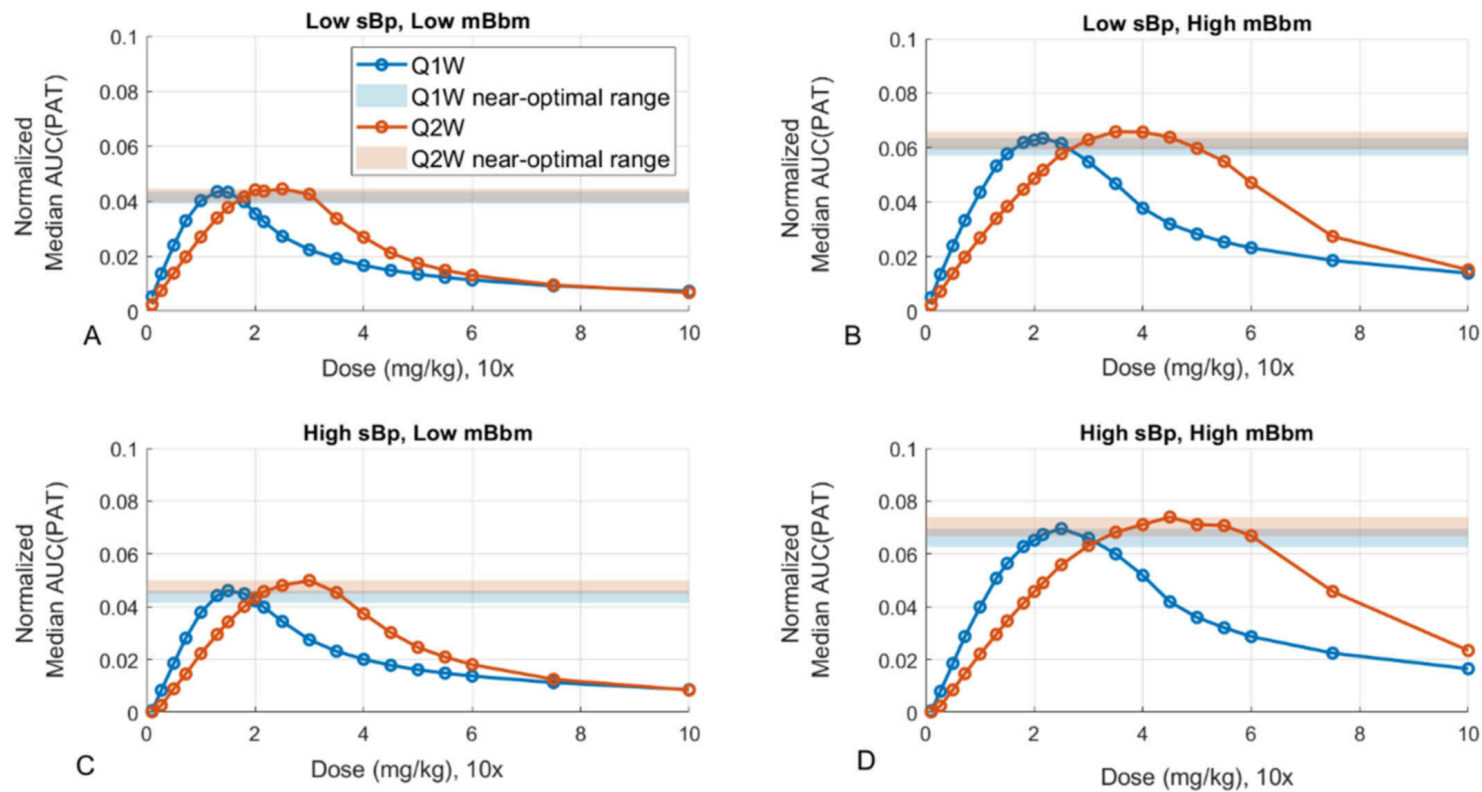


Figure 6: Patient stratification based on baseline levels of both soluble BCMA in the plasma (sBp), and membrane-bound BCMA in the TME (); high versus